

VLSI Devices

Lecture 19

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L19

Case study: TSMC 3 nm node

- IEDM 2022 (27.1)

– DIBL versus L_g

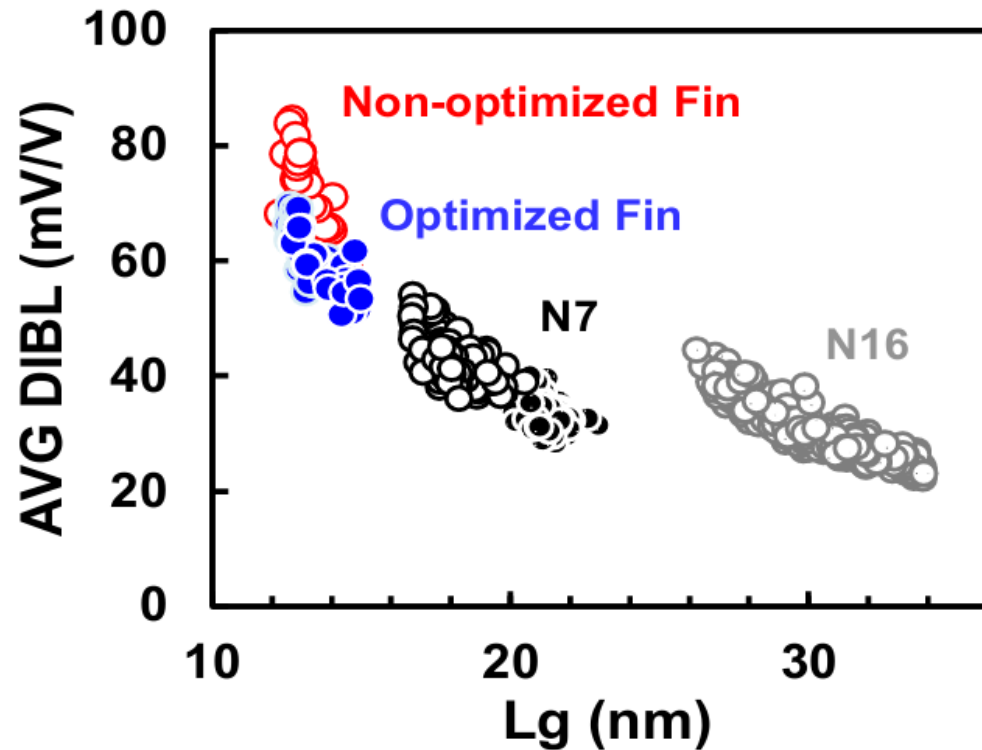


Fig.1 FinFET L_g scaling trend vs DIBL. Fin profile optimization is critical but is at the limit for further L_g scaling.

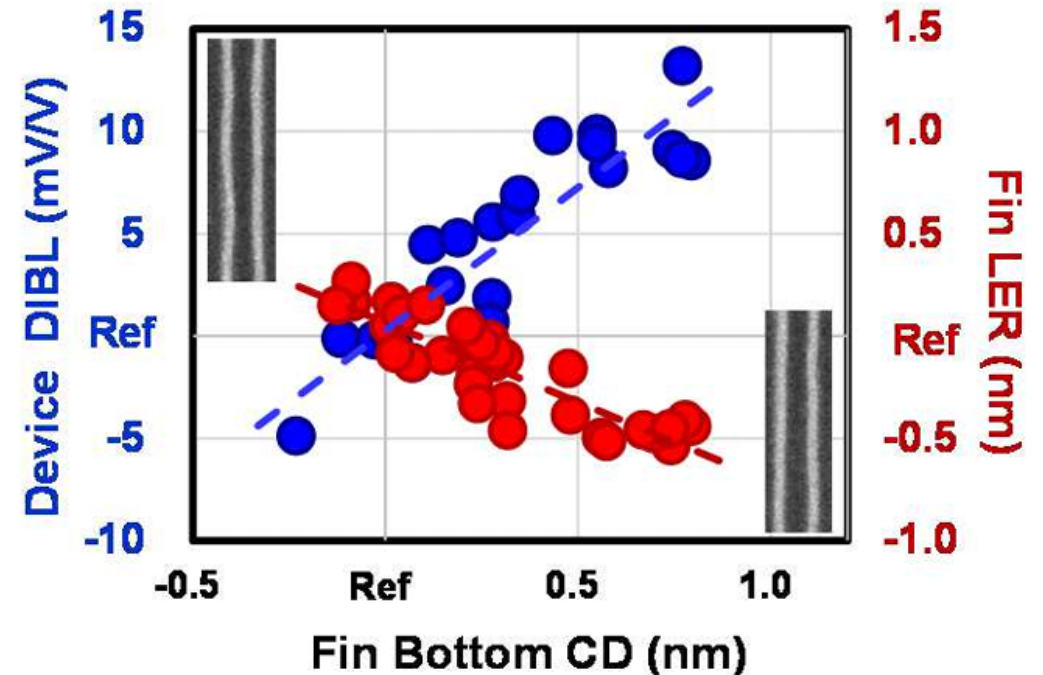


Fig.2 Smaller fin bottom CD reduces DIBL but degrades LER, which is an indicator of fin structural robustness and potential yield impact.

Source-drain series resistance

- Finite silicon resistivity + metal contact resistance
 - MOSFET channel resistance in the linear region

$$R_{ch} = \frac{V_{ds}}{I_d} = \frac{L}{\mu_{eff} C_{inv} W (V_{gs} - V_{on})}$$

Taur, ~Eq. (8.15)

Scaled contacted gate pitch

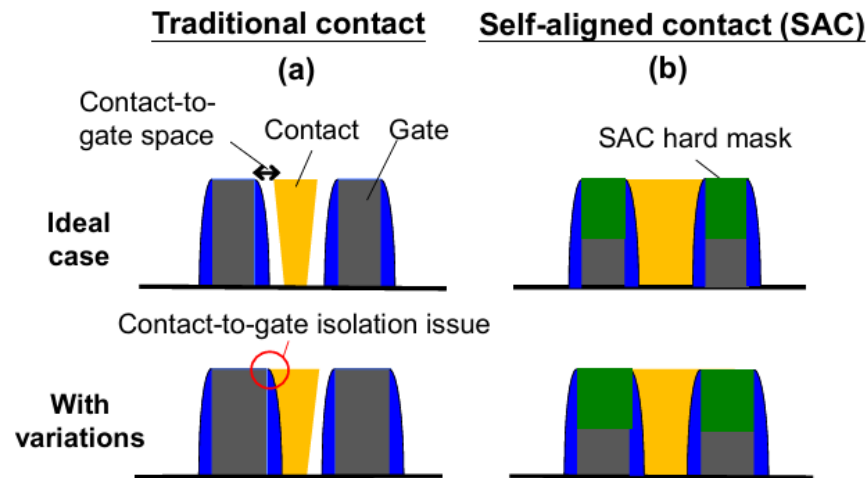
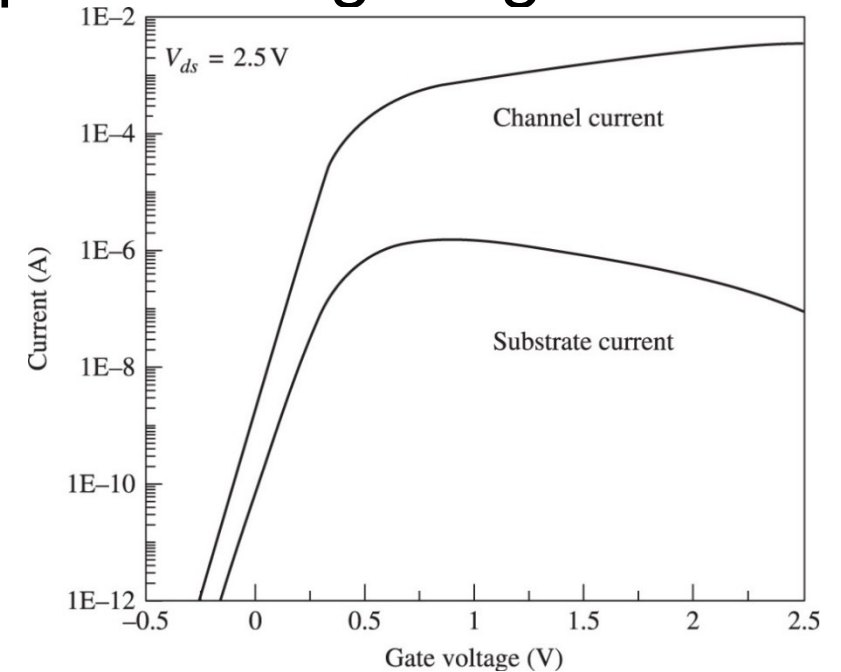
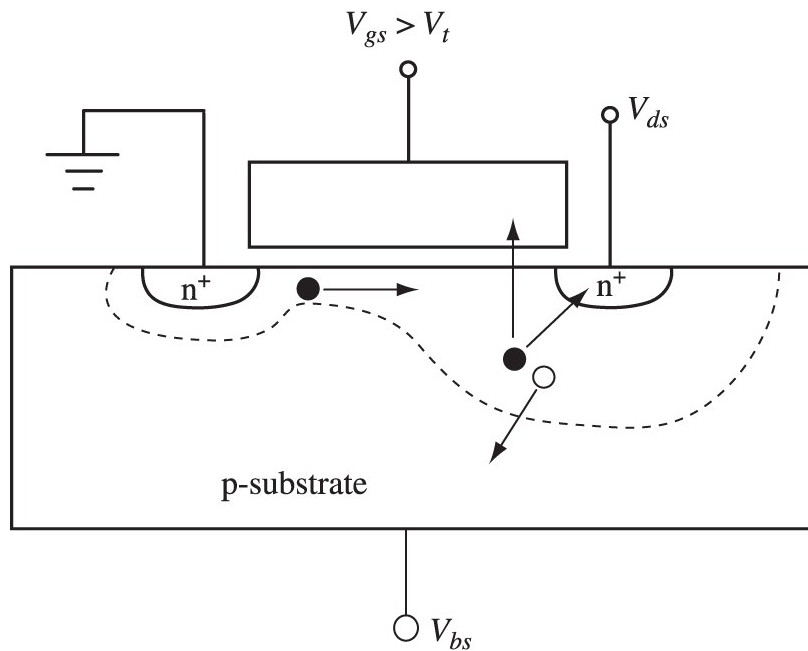


Fig.3 Contact schematics. Traditional contact (a) is vulnerable to variations induced contact-to-gate isolation issues compared to SAC (b).

Self-aligned contact (TSMC, IEDM 2022)

Hot carrier effects

- Electrons gain energy from the electric field.
 - Injection into the SiO_2 layer. Secondary electron/hole pair.
 - $V_{ds} - V_{dsat}$ is the voltage drop in the space-charge region.

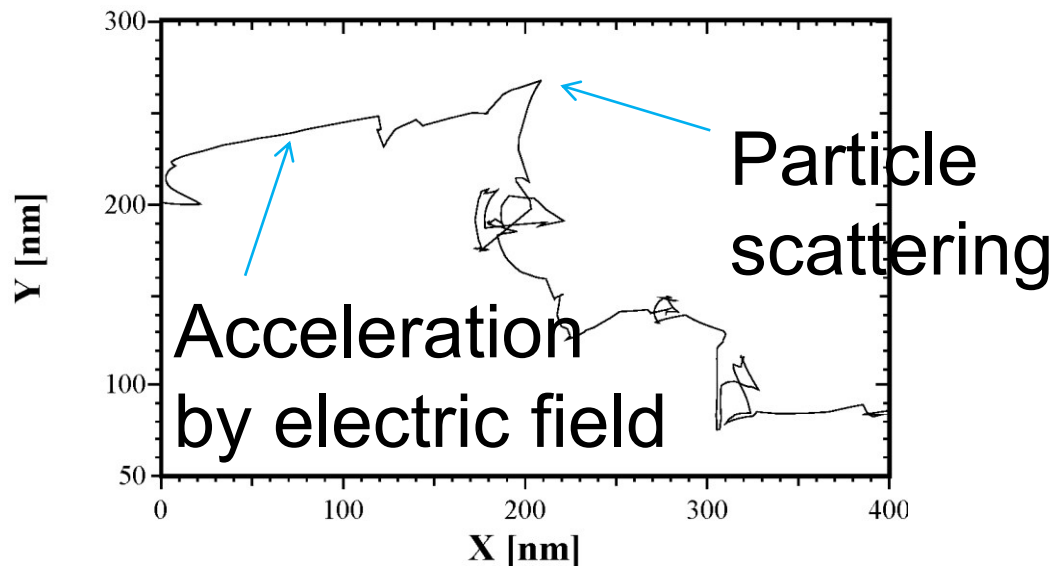


Channel hot-electron (Taur, Fig. 6.30)

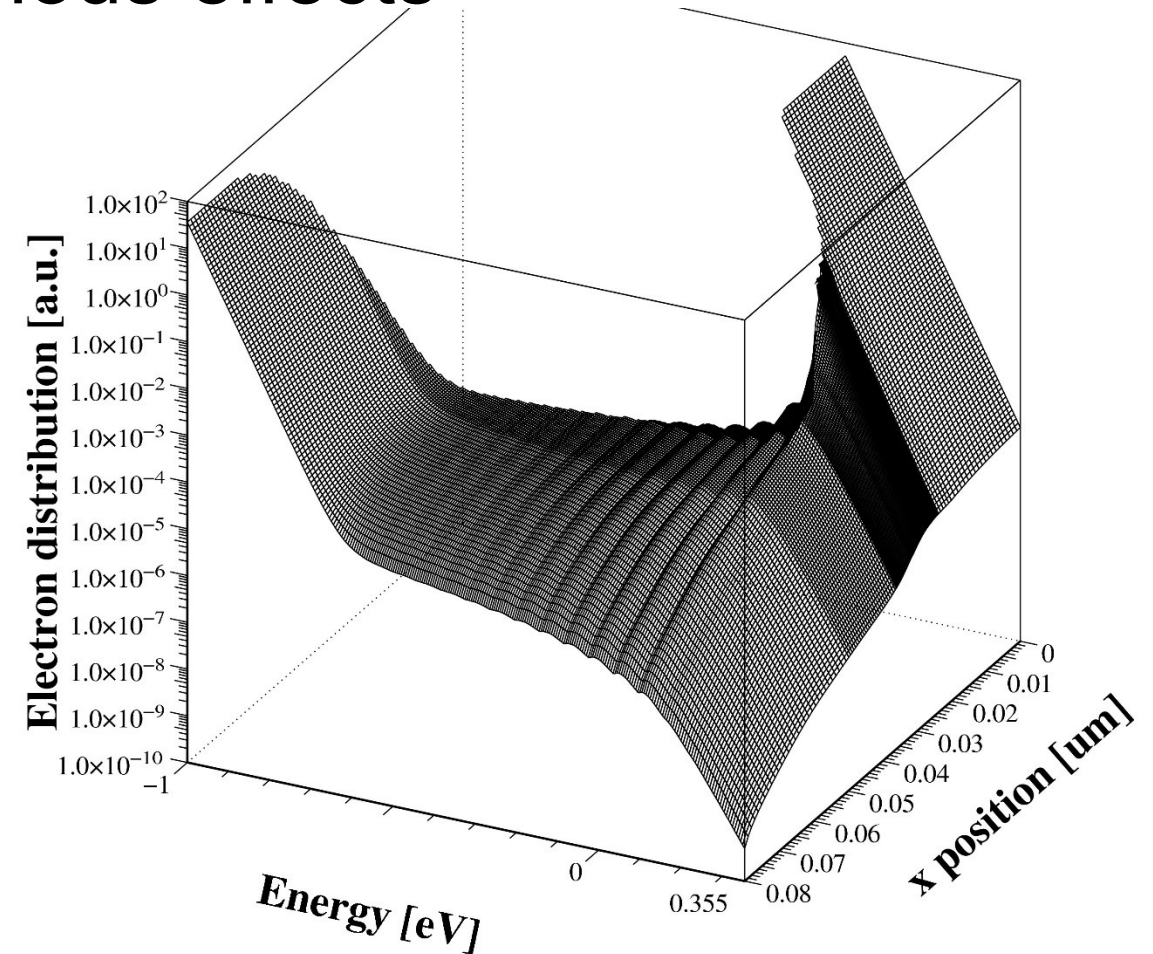
Substrate current (Taur, Fig. 6.31)

Energy distribution, $f(\mathbf{r}, E)$

- Key quantity to understand various effects
 - Various ways to model it
 - Boltzmann transport equation



Stochastic electron motion
simulated by Monte Carlo

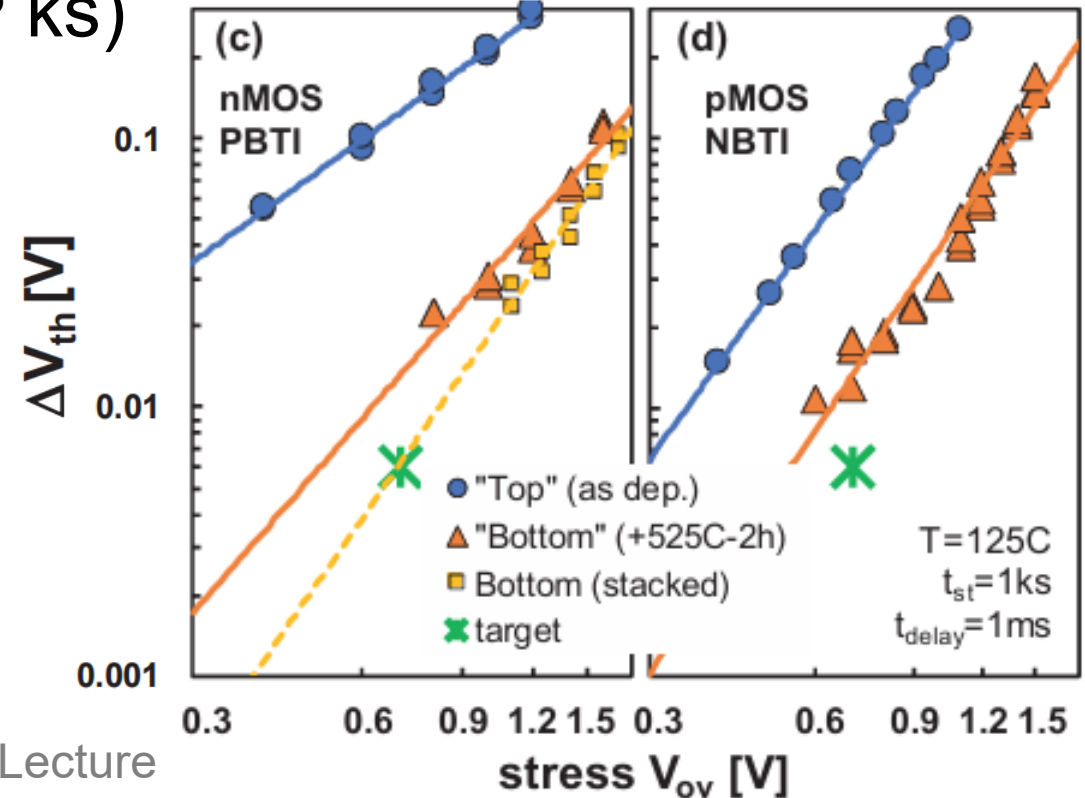


Distribution function

BTI (Bias-Temperature Instability)

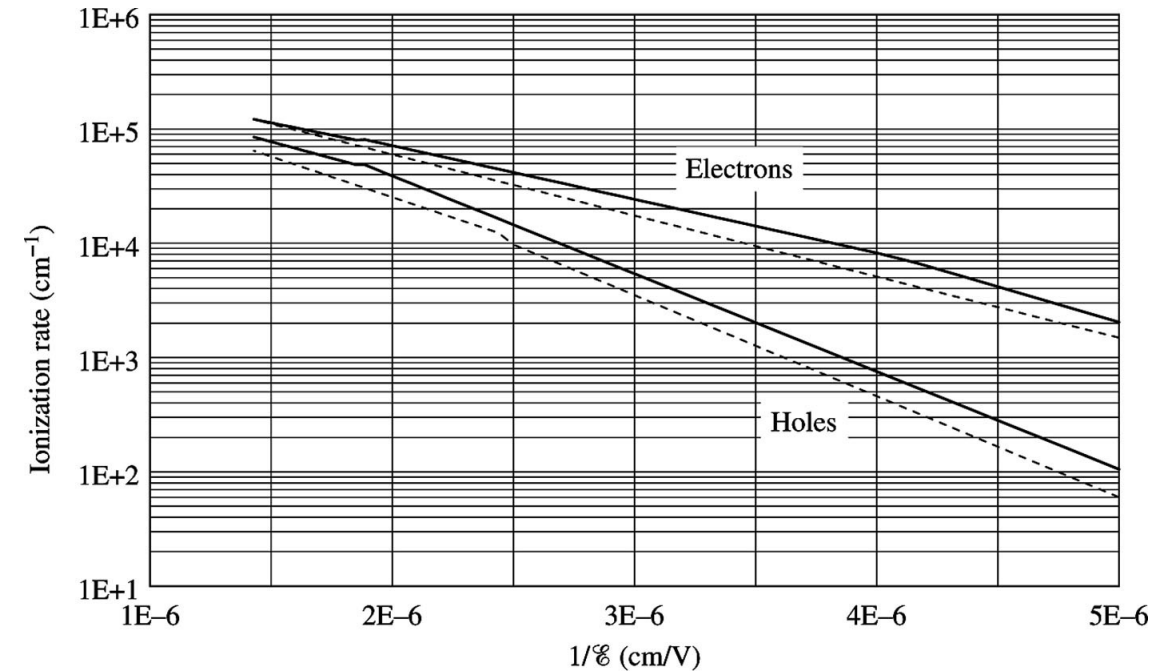
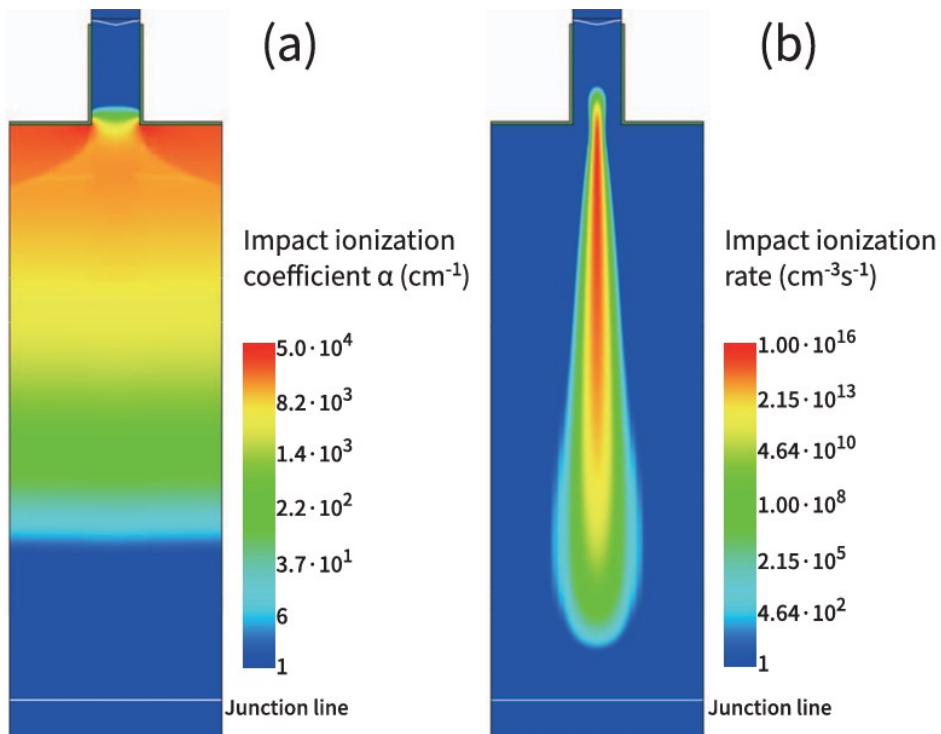
- Reliability issue (NBTI in PMOS, PBTI in NMOS)
 - Interface trap generation & $|V_t|$ shift
 - $\Delta V_{th} < 6$ mV at $V_{ov} = 0.7$ V, $T = 125$ °C, $t_{st} = 1$ ks, rescaled from 50mV at 10 years (3.16×10^5 ks)

BTI characteristics of gate stacks (IMEC, IEDM 2018)



MOSFET breakdown

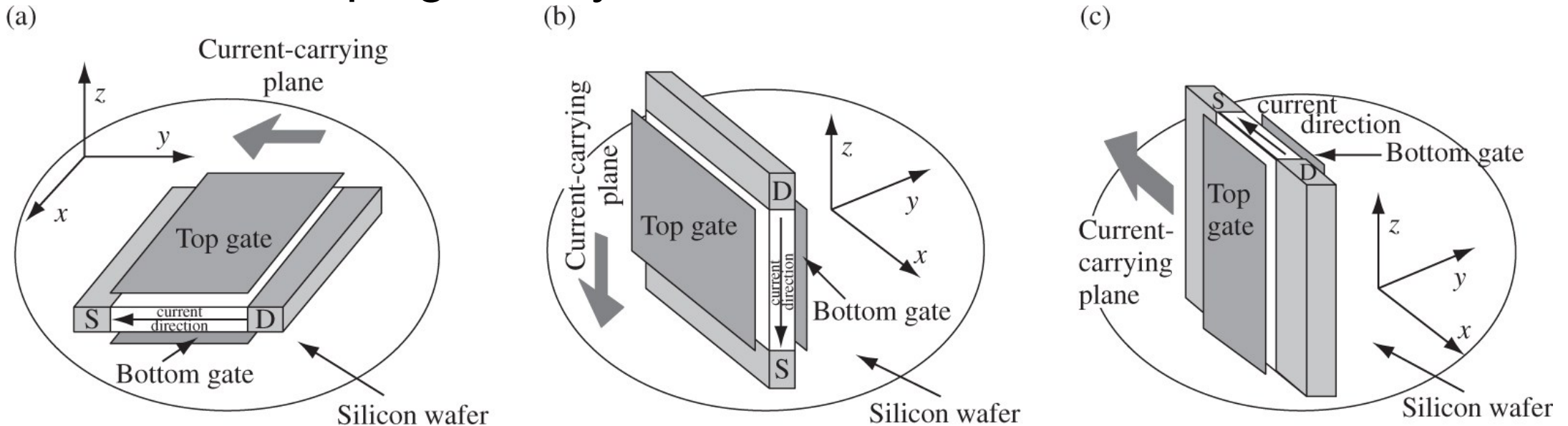
- Impact ionization
 - Strong dependence on ε
- $$\alpha = A \exp\left(-\frac{b}{\varepsilon}\right) \text{ Taur, Eq. (3.140)}$$



Impact ionization rates
in silicon (Taur, Fig. 3.23)

Double-gate (DG) MOSFETs

- Advantages of DG MOSFETs
 - Ideal 60 mV/dec subthreshold swing
 - Shorter channel length allowed
 - Channel doping is very low.



Three different orientations of double-gate MOSFETs (Taur, Fig. 7.11)

Thank you!